

EEE412 Microwave Electronics-Homework Assignment 1

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“I completed this homework assignment independently.”

Part 1

Design a third-order Chebychev (0.2 dB) LC-bandpass filter with a center frequency of f_0 and a fractional bandwidth of Δf . Choose f_0 randomly between 50 MHz and 100 MHz and $\Delta f/f_0$ between 0.1 and 0.2. The source and load impedances are both 50Ω .

I select the values as below:

$$f_0 = 85 \text{ MHz}$$

$$\Delta f = 15 \text{ MHz}$$

$$\Delta f/f_0 = 0.176$$

- a) Verify your filter design using ADS. Plot S11 and S21 in dB as a function of frequency (from $f_0 - 2\Delta f$ to $f_0 + 2\Delta f$). Note that for a good 0.2dB Chebychev filter, S11 should be less than -12dB , and S21 should be no worse than -0.2dB within the band.

n	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
3	1.228	1.153	1.228					
4	1.303	1.284	1.976	0.8468				
5	1.339	1.337	2.166	1.337	1.339			
6	1.360	1.363	2.239	1.456	2.097	0.8838		
7	1.372	1.378	2.276	1.500	2.276	1.378	1.372	
8	1.380	1.388	2.296	1.522	2.341	1.493	2.135	0.8972

Table 5.3: Table of prototype element values in Chebyshev low-pass-filters with passband ripple of 0.2 dB ($\epsilon = 0.2171$).

From the table 5.3 we are given in the filter design document we select $n=3$. We will take the values b_1 and b_3 and therefore calculate C_1 and C_3 values accordingly. Cut-off frequency will be equal to as below (see filter design pdf page 7):

$$f_c = \Delta f = 15 \text{ MHz}$$

$$C_{1,3} = \frac{b_{1,3}}{z_0 \times 2\pi \times f_c} = \frac{1.228}{50 \times 2\pi \times 15 \times 10^6} \approx 260.59 \text{ pF}$$

$$L_2 = \frac{z_0 \times b_2}{2\pi \times f_c} = \frac{50 \times 1.153}{2\pi \times 15 \times 10^6} \approx 611.69 \text{ nH}$$

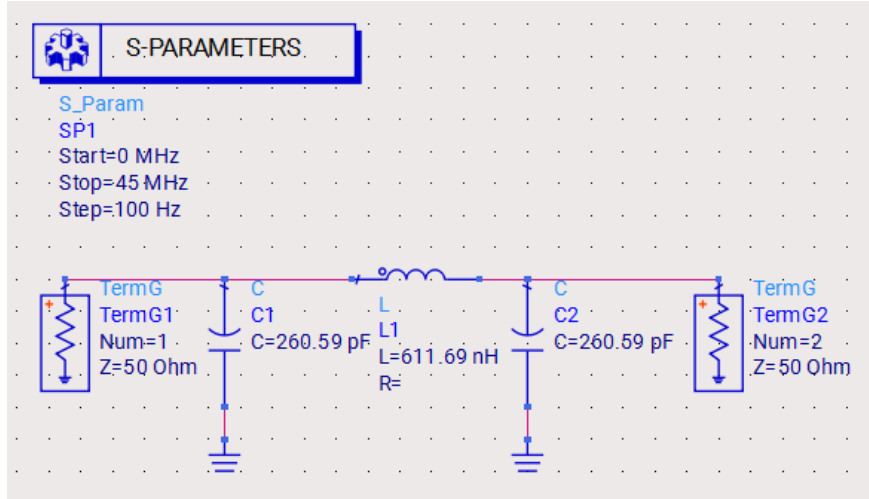


Figure 1: Initial Low-pass Filter Design Before Bandpass

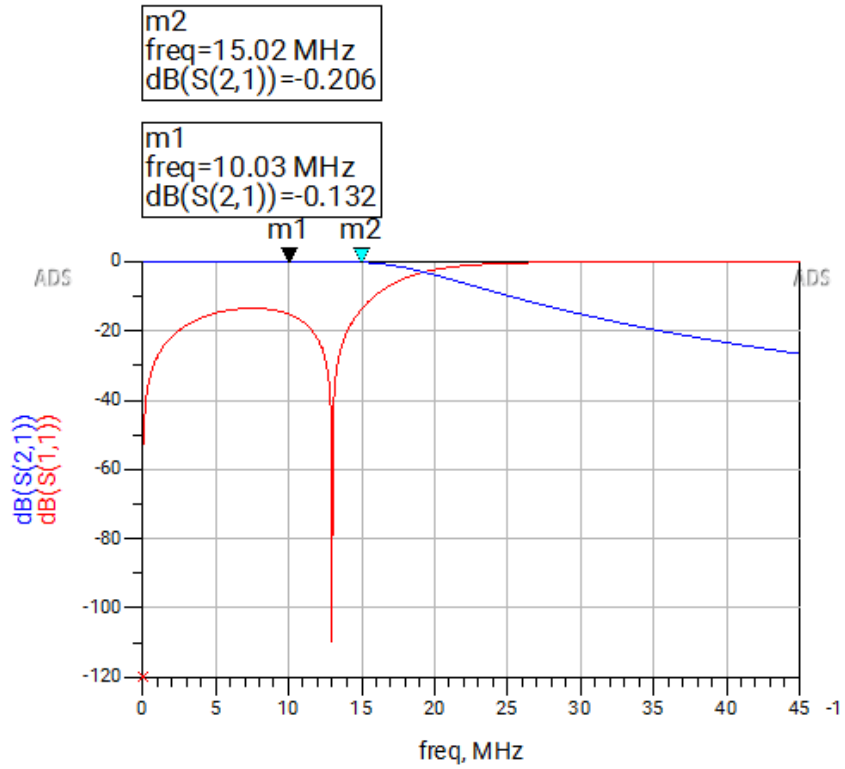


Figure 2: Plot of Figure 1

By adding a resonant couple for every inductive and conductive element at cut-off frequency we will make a bandpass filter (see eq. 5.26 and eq. 5.27 in filter design pdf).

$$C_i = \frac{1}{(2\pi \times f_0)^2 L_i} \quad C_2 = \frac{1}{(2\pi \times 85 \times 10^6)^2 \times 611.69 \times 10^{-9}} \approx 5.73 \text{ pF}$$

$$L_i = \frac{1}{(2\pi \times f_0)^2 C_i} \quad L_{1,3} = \frac{1}{(2\pi \times 85 \times 10^6)^2 \times 260.59 \times 10^{-12}} \approx 13.45 \text{ nH}$$

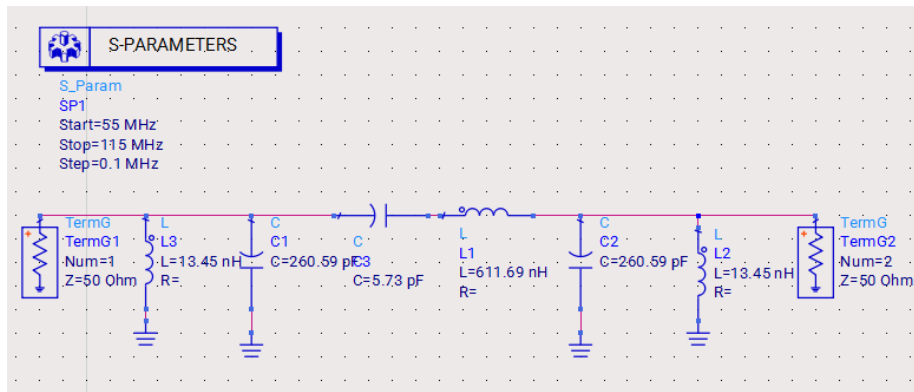


Figure 3: Ideal Filter Circuit

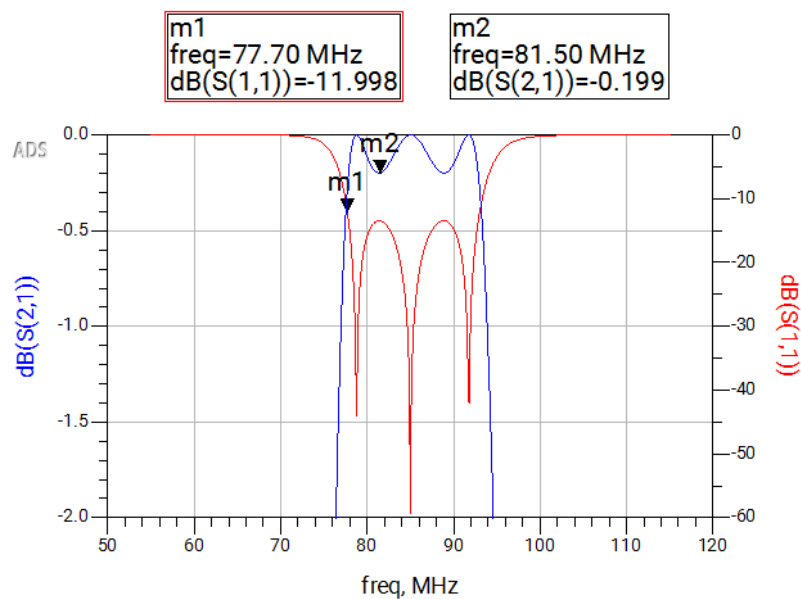


Figure 4: Ideal response: S11 and S21 (dB) vs. frequency

In the plot it is seen that the ripple is no worse than 0.2 dB and the impedance matching of the filter is close to -12 dB.

- b) Assume that the inductors have $Q=80$. Simulate your filter and plot the S11 and S21 response.

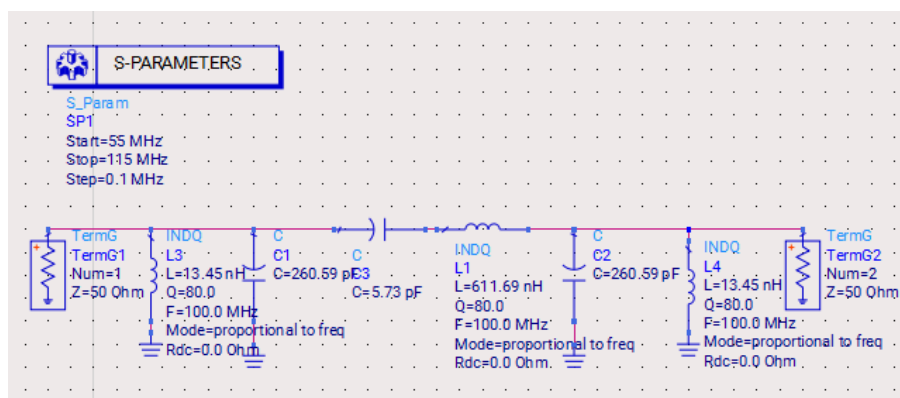


Figure 5: Filter Circuit with Q of the Inductors

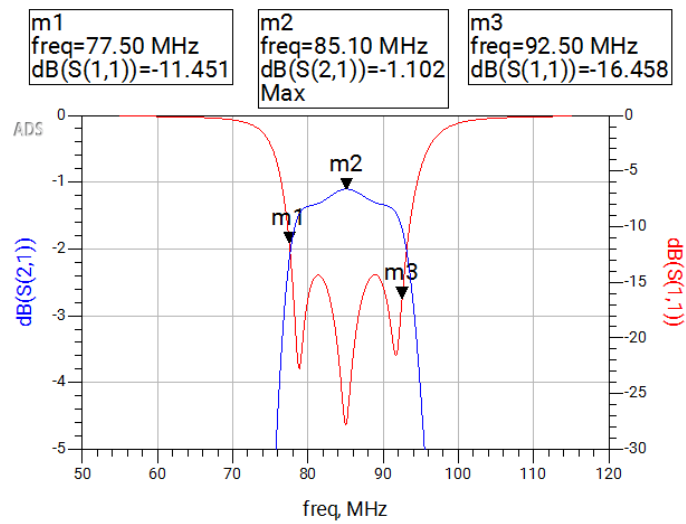


Figure 5: Response with inductor $Q = 80$

- c) Assume that there is a parasitic capacitance of 100 fF from *every* node to the ground. Find out and plot how the S11 and S21 responses are degraded because of parasitic capacitors.

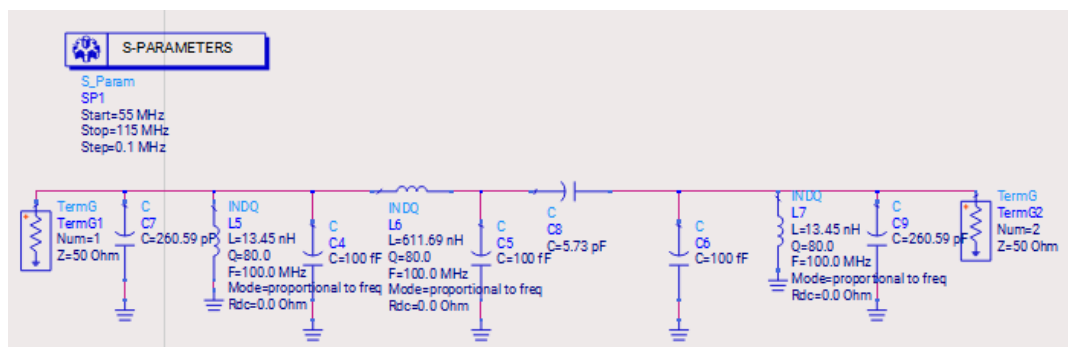


Figure 6: Filter Circuit with Q of the Inductors and Parasitic Capacitances

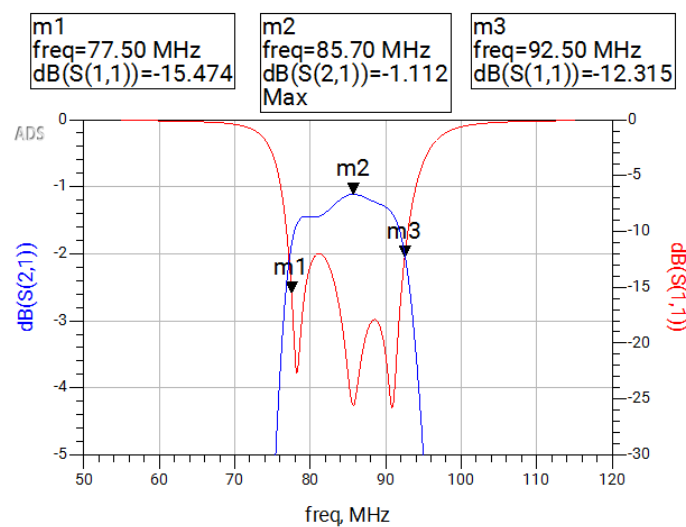
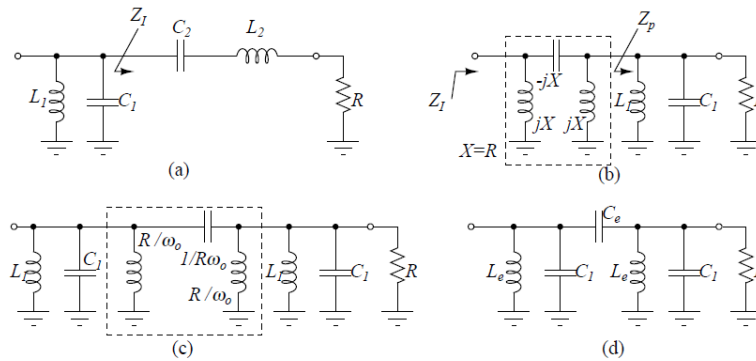


Figure 7: Degradation Due to Node Parasitic Capacitors

Since the capacitances are assumed ideal, the gain is essentially unaffected; however, the shift to the left modifies the input impedance at all frequencies and thus reduces the matching quality.

- d) Transform your filter structure so that there are no series LC branches. With all nodes containing a shunt capacitor, the parasitic capacitor of 100 fF can be used as a part of the shunt capacitance. Plot the S11 and S21 response of the modified filter.

To avoid the performance limitation of series LC branches in filters, an inverter can be utilized to convert them to parallel LC branches if the filter has a relatively small bandwidth. Again from the filter design pdf, we use the below methodology:



The network is set to operate at $f_0=85$ MHz in order to keep the matching circuit unchanged from the previous design and to maintain a 50Ω impedance at the center frequency.

$$Q = 2\pi fL/R, \quad R = 2\pi fL/Q$$

$$R_{1,3} = 2\pi \times 85 \times 10^6 \times 13.45 \times 10^{-9}/80 \approx 89.79 \text{ m}\Omega \quad R_2 = 2\pi \times 85 \times 10^6 \times 611.69 \times 10^{-9}/80 \approx 4.084 \Omega$$

Handwritten calculations and circuit diagrams:

Circuit diagram 1: A series LC branch with inductor L_1 and capacitor C_1 in series, followed by a parallel LC branch with inductor L_2 and capacitor C_2 in parallel, connected to a load resistor $R_L = 50 \Omega$. The input impedance is Z_I and the output impedance is Z_P .

Calculation 1: $Z_1 = \left(R_L \parallel \frac{1}{j\omega C_2} \parallel (j\omega L_3 + R_{S3}) \right) \approx R_L = 50 \Omega$ at 85 MHz

Calculation 2: $Z_2 = Z_1 + \frac{1}{j\omega C_2} + j\omega L_2 + R_{S2}$

Calculation 3: $Z_2 \approx R_{S2} + Z_1 \approx 54.084 \Omega$ at 85 MHz

Calculation 4: $Z = \frac{Z_2^2}{Z_1}$ and $Z = Z_2 \approx 54.084 \Omega$ desired

Calculation 5: therefore, $X = \sqrt{Z_1 Z_2} = \sqrt{54.084 \times 50} \approx 52.002 \Omega$

Calculation 6: so C values $\rightarrow \frac{1}{\omega C} = \frac{1}{2\pi \times 85 \times 10^6 \times C} \approx 52$

Calculation 7: $C = 3.01 \text{ pF}$

Calculation 8: L value $= \omega L = 2\pi \times 85 \times 10^6 \times L = 52$

Calculation 9: $L = 37.37 \text{ nH}$

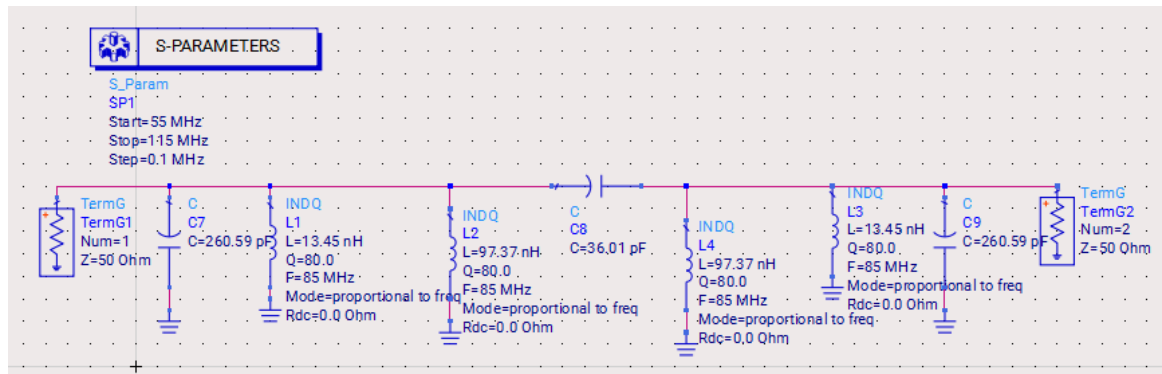


Figure 8: Transformed Filter Topology After Calculations

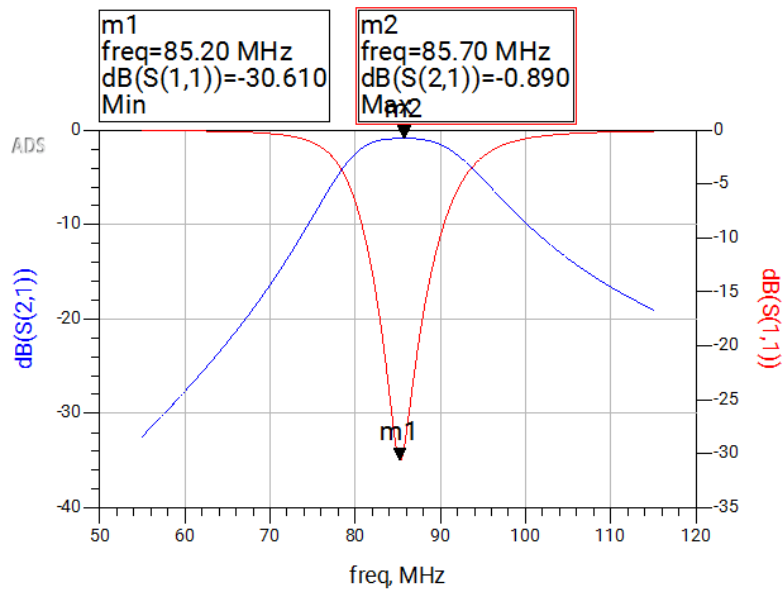
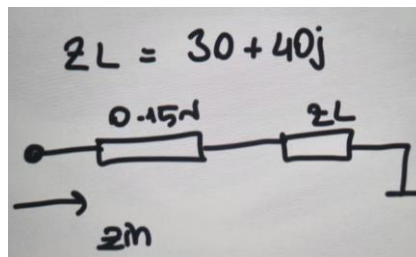


Figure 9: Plot of the Transformed Topology

Part 2

A 0.15λ long transmission line of characteristic impedance $Z_0=50\Omega$ is terminated with a load impedance of $Z_L=R+jX$ at 3 GHz. (Choose random R and X values between 10Ω and 250Ω).



- a) Find the impedance at the other end of the line using the relevant formula.

$$\beta L = \frac{2\pi}{\lambda} \times 0.15$$

$$= 0.3\pi$$

$$\approx 54^\circ$$

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta L)}{Z_0 + jZ_L \tan(\beta L)}$$

$$Z_{in} = 50 \frac{30 + j40 + j50 \tan(54^\circ)}{50 + j(30 + j40) \tan(54^\circ)}$$

$$Z_{in} = 2.508835 - 1.033697j \text{ (normalized)}$$

- b) Find the value of the input impedance using the Smith Chart. Show your work on the Smith chart.

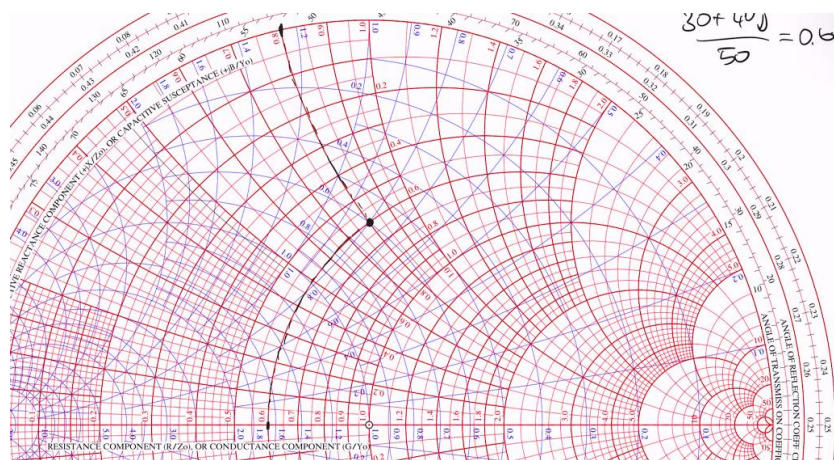


Figure 10: $Z_{in} = 0.6 + j0.8$ is found in the Smith Chart

The point is subsequently turned clockwise by 0.15λ from the origin, preserving its distance. The new position matches the previously calculated point.

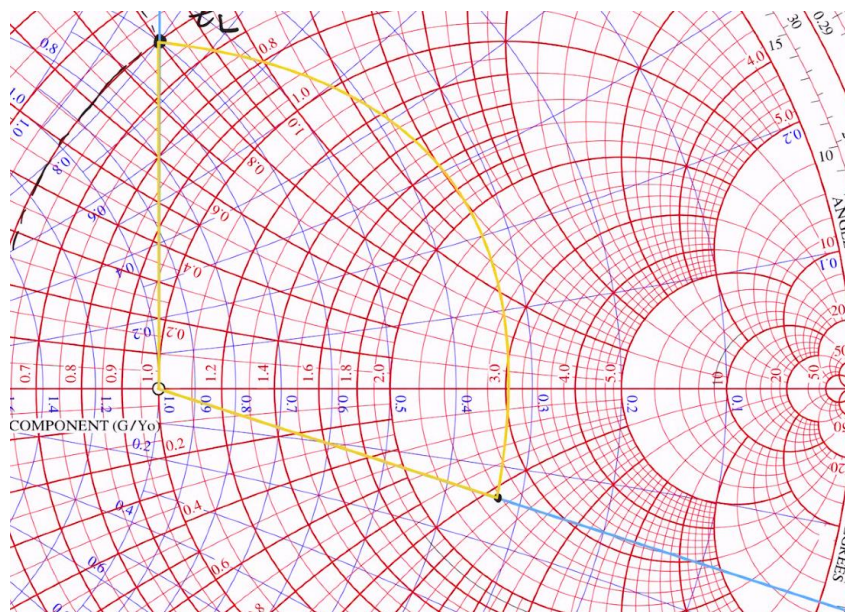


Figure 11: $Z_{in} = 2.5088 - j1.033697$ is found in the Smith Chart

c) Verify your result in a) and b) using ADS with $f=3$ GHz. You may use a transmission line element with an electrical length of 54 degrees at 3 GHz.

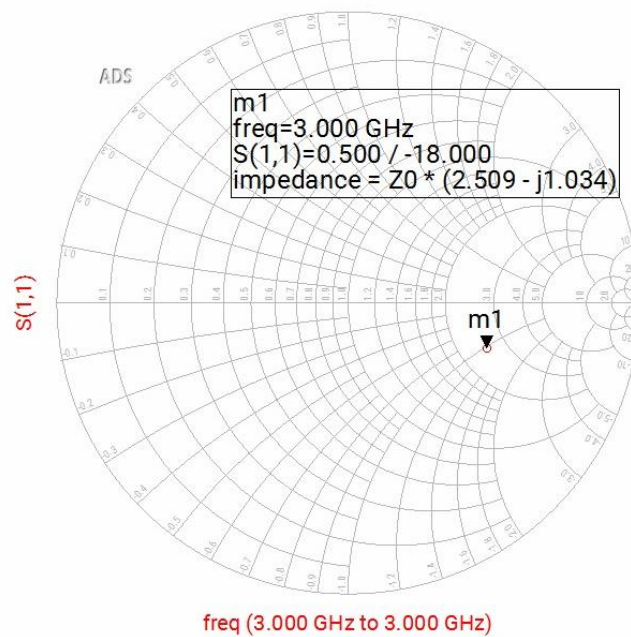


Figure 12: ADS verification of Z_{in} for 0.15λ

Seen that above calculations are correct.

d) Find the approximate transmission line length in λ units (instead of 0.15λ) to make the input impedance purely real using the Smith chart.

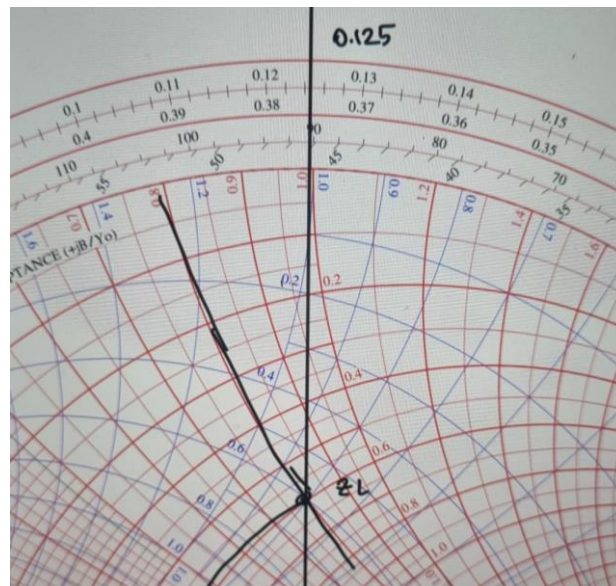


Figure 13: 0.125λ found from Smith Chart

- $0.25 \lambda - 0.125 \lambda = 0.125 \lambda$

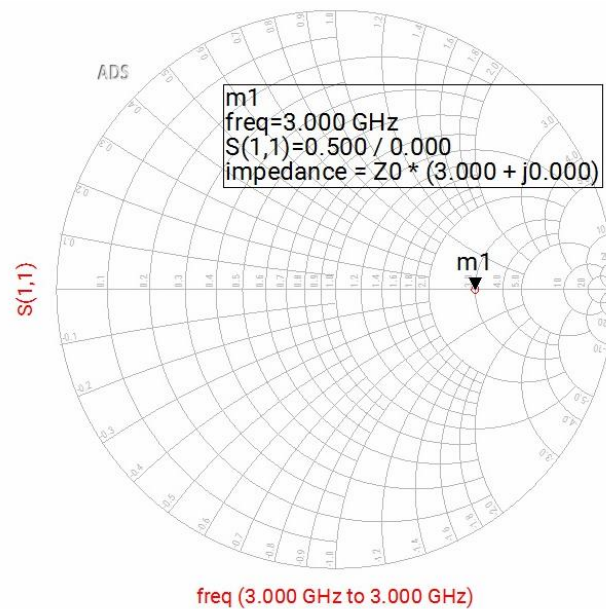
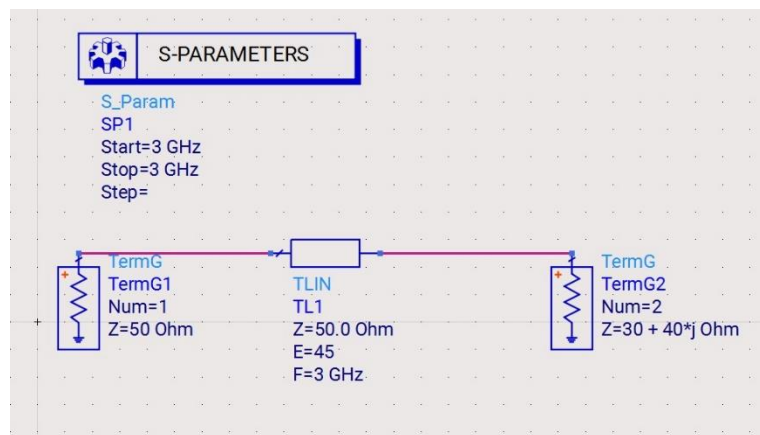
- $0.5 \lambda - 0.125 \lambda = 0.375 \lambda$

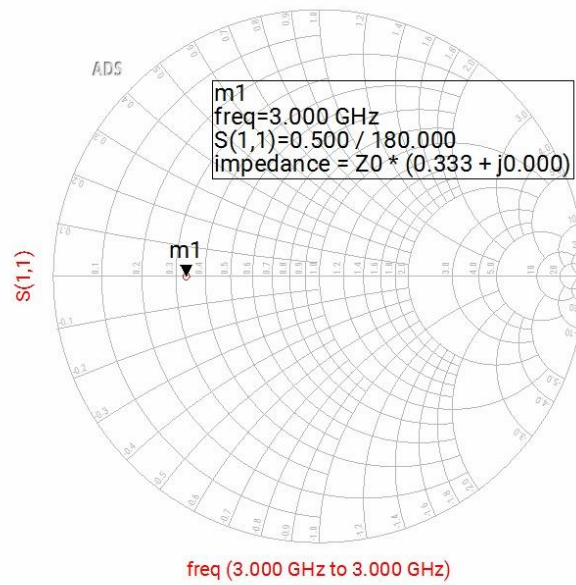
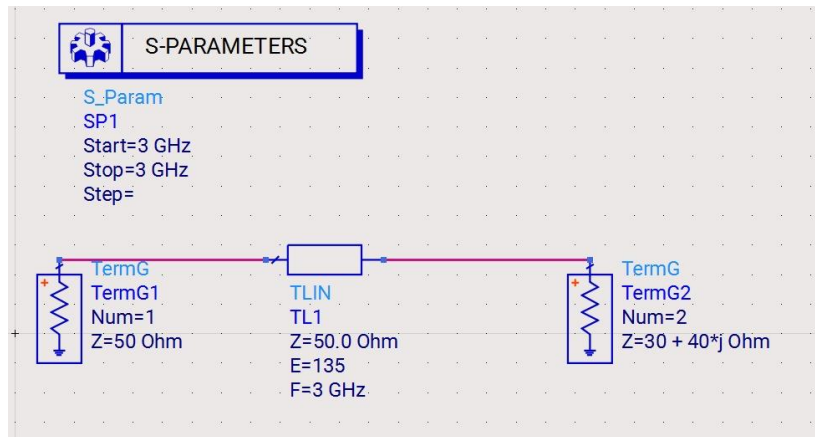
Then the electrical length of them is found :

$$2 \cdot 0.375 \cdot 180 = 135$$

$$2 \cdot 0.125 \cdot 180 = 45$$

e) Verify your result in d) using ADS at $f=3$ GHz. Note that you can use the real and imaginary parts of the Z_{11} of the network to find the input impedance.





A comparison of Figure 11 and 13 with the ADS results again shows consistency: the results are correct, and the purely real impedances agree with the calculated ones.

References

1. Course handout and assignment text (HW#1, EE 412/512).
2. Course handout (FilterDesign, EE 412/512).
3. Desmos Scientific Calculator